Deep Equilibrium Convolutional Sparse Coding for Hyperspectral Image Denoising

Jin Ye, Jingran Wang, Fengchao Xiong, *Member, IEEE*, Jingzhou Chen, *Member, IEEE*, and Yuntao Qian, *Senior Member, IEEE*

***Abstract*—Hyperspectral images (HSIs) play a crucial role in remote sensing but are often degraded by complex noise patterns. Ensuring the physical property of the denoised HSIs is vital for robust HSI denoising, giving the rise of deep unfolding-based methods. However, these methods map the optimization of a physical model to a learnable network with a predefined depth, which lacks convergence guarantees. In contrast, deep equilibrium (DEQ) models treat the hidden layers of deep networks as the solution to a fixed-point problem and models them as infinite-depth networks, naturally consistent with the optimization. Under the framework of DEQ, we propose a deep equilibrium convolutional sparse coding (DECSC) framework that unifies local spatial–spectral correlations, nonlocal spatial self-similarities, and global spatial consistency for robust HSI denoising. Within the convolutional sparse coding (CSC) framework, we enforce shared 2-D convolutional sparse representation to ensure global spatial consistency across bands, while unshared 3-D convolutional sparse representation captures local spatial–spectral details. To further exploit nonlocal self-similarities, a transformer block is embedded after the 2-D CSC. In addition, a detail enhancement module is integrated with the 3-D CSC to promote image detail preservation. We formulate the proximal gradient descent of the CSC model as a fixed-point problem and transform the iterative updates into a learnable network architecture within the framework of DEQ. Experimental results demonstrate that our DECSC method achieves superior denoising performance compared to state-of-the-art methods.**

***Index Terms*—Convolutional sparse coding (CSC), deep equilibrium (DEQ) model, hyperspectral image (HSI) denoising.**

NOMENCLATURE

X	Clean hyperspectral image (HSI).
Y	Noisy HSI.
N	Additive noise.
C	Global interband common (GIC) structures.
U	Local spatial–spectral unique (LSU) structures.
$\mathbf{K}_b$	2-D convolutional dictionary for the bth band of the HSI.
S	3-D Corresponding sparse coefficients of **C**.
D	convolutional dictionary of **U**.
H	Corresponding sparse coefficients of **U**.
α^t	Result of the tth iteration of the deep equilibrium (DEQ).
α^*	Equilibrium point of DEQ.
z	Noisy image injected into each layer of DEQ.

Received 3 June 2025; revised 14 August 2025; accepted 25 August 2025. Date of publication 27 August 2025; date of current version 4 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62371237, in part by the Fundamental Research Funds for the Central Universities under Grant 30923010213 and Grant 30925020213, and in part by Jiangsu Provincial Natural Science Foundation of China under Grant BK20241454. *(Corresponding author: Fengchao Xiong.)*

Jin Ye, Jingran Wang, and Jingzhou Chen are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

Fengchao Xiong is with the School of Computer Science and Engineering and the State Key Laboratory of Intelligent Manufacturing of Advanced Construction Machinery, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: fcxiong@njust.edu.cn).

Yuntao Qian is with the College of Computer Science, Zhejiang University, Hangzhou 310027, China.

Digital Object Identifier 10.1109/TGRS.2025.3603390

I. INTRODUCTION

HSI, with its rich spectral information, is widely used in remote sensing [1], medical diagnostics [2], agriculture [3], and object tracking [4], [5], [6]. Increasing spectral resolution comes at the cost of reducing the number of photons received in each channel. Combined with atmospheric interference, this inevitably introduces noise during the sensing process, which can degrade subsequent applications. Therefore, HSI denoising is a crucial preprocessing step to ensure image quality and enable reliable downstream applications.

The inherent spatial and spectral redundancy in HSIs allows clean HSIs to be represented as the combinations of few elements from a dictionary, which has driven the success of sparse coding in HSI denoising. Due to the high dimensionality of reshaping the whole image into a vector, traditional sparse coding model typically requires partitioning the whole HSIs into multiple overlapping patches, thereby ignoring the shift-invariant properties of the data. As an alternative, convolutional sparse coding (CSC) employs a set of convolutional atoms to represent the image, naturally preserving the spatial relationships between pixels. Owing to this advantage, CSC has been widely applied to various inverse problems [7], [8], [9]. In the context of HSI denoising, Xiong et al. [10] and Yin and Chen [11] extended 2-D CSC models to 3-D ones by employing 3-D convolutions to jointly model local spatial–spectral correlations but fail to capture the global spectral correlations among bands. More recently, Tu et al. [12] enforced shared convolutional sparse coefficients across bands to capture interband spatial structural consistency. Nevertheless, this approach overlooks the local spatial–spectral correlations.

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

In addition, the CSC model is based on the hand-crafted sparsity prior and cannot enjoy the data-driven learning from data. To address this limitation, the deep unfolding technique has been used in [10] and [12] to transform the iterative optimization process into a deep neural network (DNN) with a fixed number of weight-tied layers, where each layer mimics one step of the original optimization. One major limitation of the deep unfolding models is the hardware memory constraints: during training, all intermediate activations must be stored for backpropagation, restricting the model depth and complicating the training. As a result, such unfolding networks are typically constrained to a small number of layers. However, the fixed number of iterations may not guarantee convergence, and running additional iterations at test time can lead to significant performance degradation [13], [14].

In contrast, DEQ models [15] offer a paradigm shift by treating the hidden layers of deep networks as the solution to a fixed-point problem. Instead of unfolding the optimization process into predefined depth, DEQs model infinite-depth networks and directly solve for the equilibrium (fixed) point during the forward pass. This fixed-point formulation makes DEQs particularly suitable for tasks involving iterative refinement, such as those frequently encountered in convex and nonconvex optimization for model-driven methods. In addition, DEQs leverage existing numerical solvers and implicit differentiation for forward evaluation and backward propagation, significantly reducing memory usage and improving training stability. In a nutshell, DEQ offers a manner to transform the traditional model-driven methods into a learnable architecture with convergence guarantees and preferable interpretability to bypass the limitation of hand-crafted priors. In the literature, Gilton et al. [13] demonstrated the improved effectiveness of DEQs in a variety of image reconstruction tasks over deep unfolding networks. Zhao et al. [16] extended the DEQ framework to recurrent neural networks (RNNs) and and plug-and-play (PnP) algorithms for snapshot compressive imaging. More recently, Geng et al. [17] applied DEQ to accelerate sampling in diffusion models.

Building upon the strengths of CSC and DEQ, we propose a novel deep equilibrium CSC (DECSC) framework for HSI denoising. To fully exploit both local spatial–spectral and global spectral correlations inherent in HSIs, we decompose the hyperspectral data into two components: 1) GIC structures and 2) LSU structures. For modeling the GIC structure, we enforce shared convolutional sparse coefficients across bands, thereby ensuring spatial consistency throughout the spectral dimension. For local structures, we employ 3-D convolution to jointly capture fine-grained spatial–spectral dependencies, preserving spatial–spectral details within nearby bands. To overcome the limitations of hand-crafted sparsity priors, we integrate a transformer module to capture the nonlocal spatial self-similarity within the GIC component. In addition, we introduce a detailed enhancement module to promote detail preservation within the LSU component. Within the DEQ framework, we reformulate the estimation of convolutional sparse coefficients as a fixed-point problem, resulting in an implicit infinite-depth learnable network that not only integrates the physical interpretability of the model but also offers guaranteed convergence properties. Extensive experiments on both synthetic and real-world datasets validate the superior denoising performance of the proposed DECSC method.

The remainder of this article is organized as follows. Section II reviews recent advances in HSI denoising and DEQ. Section III details the proposed DECSC, including its network architecture and optimization under the DEQ framework. Section IV presents the extensive experimental results validating our approach. Finally, Section V concludes this article.

II. Related Works

This section presents a concise review of recent advancements in HSI denoising and DEQ.

A. HSI Denoising

In recent years, the paradigm of HSI denoising has shifted from model-driven-to-data-driven methods, with hybrid-driven approaches emerging as a promising direction that integrates the advantages of both. Defining an effective prior is crucial in traditional model-based methods. The strong spatial and spectral correlations in HSIs enable their representation in a low-dimensional subspace, underpinning the success of low rankness and sparsity priors [18], [19]. Methods based on low-rank matrix or tensor decomposition [20], [21] and rank minimization [22], [23], [24] denoise HSIs by extracting these low-rank structures. Meanwhile, the intrinsic redundancy of HSIs supports sparse representations over a few atoms from a dictionary. Peng et al. [25] reconstructed clean HSIs using either predefined or learned dictionaries, and Zhao and Yang [26] employed sparse coding to capture both global spatial and local spectral redundancies. Furthermore, Zhuang et al. [27] preserved "rare pixels" containing critical information through collaborative sparsity. Total variation (TV) regularization—another form of transformed sparsity—has also proven effective to promote piecewise smoothness [28]. Despite their interpretability and effectiveness, these model-driven approaches often demand extensive parameter tuning and entail high computational costs.

Data-driven methods utilize various DNN architectures to model the intrinsic structure of HSI from data. Convolutional neural networks (CNNs) [29], [30] can only model the local spatial–spectral correlation with limited receptive fields, restricting their ability to capture long-range dependencies that are always important in image processing. Transformers treat nonoverlapping image patches as sequences and leverage attention mechanisms to model long-range dependencies [31]. Compared with RGB images, HSIs possess significantly richer spectral resolution and exhibit unique spatial–spectral characteristics. Consequently, transformer-based methods for HSI denoising often emphasize the exploration of both spatial and spectral self-similarity. For instance, Peng et al. [25] employed a dual-branch network combining CNNs and transformers to separately capture spectral correlations and nonlocal spatial self-similarity. Li et al. [32] adopted nonlocal spatial self-attention and global spectral attention to fully exploit the inherent similarities along spatial and spectral dimensions. More recently, some approaches have combined self-attention

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

with carefully designed low-rank modules to simultaneously capture spatial self-similarity and spectral low-rank structure [33], [34], [35]. However, the quadratic complexity of transformers in handling long sequences leads to substantial computational overhead, limiting their scalability and practicality in HSI applications. To address this, recent works have explored Mamba [36], a state space model architecture, for HSI denoising [37]. Data-driven neural networks often require extensive empirical tuning and trial-and-error architecture design and cannot share the well-established interpretability of model-based methods.

Hybrid-driven methods have emerged as a promising research direction by combining the interpretability of model-driven approaches with the strong representation capabilities of data-driven techniques [38], [39]. To incorporate low-rank priors in HSIs, several studies [40], [41], [42], [43] have projected HSIs into low-dimensional spectral subspaces to facilitate efficient restoration. Similarly, sparsity priors have inspired the design of network architectures based on deep unfolding [10], [44], [45], [46], where the network structure is derived by unfolding the optimization process of a model. For instance, Xiong et al. [47] converted the iterative soft-shrinkage algorithm of a multitask sparse representation model into an explainable network for enhanced HSI denoising. In [48], a TV-based denoising model was unfolded into a learnable network. By further integrating a statistical feature injection module and a multiscale degradation guidance module, the method effectively recovers real structural details. Despite these advantages, deep unfolding networks often suffer from high memory consumption, instability, and numerical issues arising in backpropagation, especially as the number of iterations increases, negatively impacting the reconstruction performance.

B. DEQ Model

Traditional neural networks typically enhance their representational capacity by increasing the number of explicit layers. However, this strategy often leads to increased memory consumption and computational overhead. Recent research has shown that comparable or even superior performance can be achieved by sharing weights across all layers [49], [50]. Motivated by this insight, Bai et al. [15] introduced the DEQ, which abandons the conventional notion of a finite stack of layers. Instead, DEQ defines an implicit infinite-depth network by formulating a set of analytical conditions whose solution corresponds to the network's output. Rather than unrolling layers, DEQ directly solves for the equilibrium point where the hidden representation becomes stable. This point can be found using efficient black-box root-finding methods such as Broyden's method or Anderson acceleration. Crucially, DEQ leverages implicit differentiation to compute gradients during backpropagation without storing intermediate activations, enabling memory usage to remain constant at $\mathcal{O}(1)$. To enhance training stability, several follow-up studies have introduced techniques such as convergence-enforcing layers [51], Jacobian regularization [52], and phantom gradients [53]. Notably, the iterative optimization process in model-based methods naturally aligns with the equilibrium-seeking principle of DEQ, making it well-suited for tasks in imaging [13], [16] and denoising [54]. Beyond optimization, the multiscale fusion can also be framed within an equilibrium formulation. For instance, Bai et al. [55] proposed the multiscale DEQ (MDEQ) model, where inputs are injected at the highest resolution and propagated implicitly across scales to satisfy a joint equilibrium condition. In summary, the DEQ model provides an attractive manner to build the connection between the fixed-point optimization and learning-based techniques to enjoy both benefits. For this end, we adopt the DEQ model to transform the optimization of the CSC model for HSI denoising.

III. Method

In this section, we first define the problem and outline the motivation behind our approach. We then detail our DECSC, including its network layer design and forward and backward strategies. The main notations used in this section are listed in the Nomenclature.

A. CSC Model for HSI Denoising

Let $\mathbf{Y} \in \mathbb{R}^{H\times W\times B}$ be an HSI with $H\times W$ pixels and B bands. Typically, the observed noisy HSI $\mathbf{Y}$ is modeled as follows:

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ is the clean HSI to be recovered and $\mathbf{N} \in \mathbb{R}^{H\times W\times B}$ represents the additive noise.

Unlike conventional RGB images, HSIs capture the spectral reflectance of each pixel across contiguous wavelengths. Accurately modeling these spatial–spectral structures is critical for effective HSI denoising. While spectral reflectance varies with wavelength, the spatial distribution of scene objects remains largely consistent across bands, resulting in globally shared spatial structures. In addition to these commonalities, HSIs also exhibit local spatial–spectral variations unique to individual bands, which encode fine details. Motivated by these observations, we decompose the clean HSI into two complementary components—GIC structures $\mathbf{C}$ and LSU structures $\mathbf{U}$

$$\mathbf{X} = \mathbf{C} + \mathbf{U}. \tag{2}$$

We leverage the success of sparse representations to capture $\mathbf{C}$ and $\mathbf{U}$. Specifically, each band of $\mathbf{C}$ is modeled as a sparse combination of convolutional atoms from a band-specific dictionary, while enforcing shared sparse coefficients across all bands to reflect common spatial structures

$$\mathbf{C}_1 = \mathbf{K}_1 \star \mathbf{S} = \sum_{m=1}^{M} \mathbf{k}_{1,m} \star \mathbf{S}_m$$
$$\vdots$$
$$\mathbf{C}_b = \mathbf{K}_b \star \mathbf{S} = \sum_{m=1}^{M} \mathbf{k}_{b,m} \star \mathbf{S}_m$$
$$\vdots$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

$$\mathbf{C}_B = \mathbf{K}_B \star \mathbf{S} = \sum_{m=1}^{M} \mathbf{k}_{B,m} \star \mathbf{S}_m \tag{3}$$

where $\star$ denotes the 2-D convolutional operator, $\mathbf{K}_b = \{\mathbf{k}_m\}_{m=1}^M$ is the 2-D dictionary for band b, and $\mathbf{S} = \{\mathbf{S}_m\}_{m=1}^M$ represents the shared sparse coding. Here, we use different 2-D filters (dictionaries) for each band to map all of them into a shared sparse subspace to encourage the global consistency.

For the local unique spatial–spectral component $\mathbf{U}$, we employ a 3-D CSC model

$$\mathbf{U} = \mathbf{D} \star \mathbf{H} = \sum_{j=1}^{J} \mathbf{d}_j \star \mathbf{h}_j \tag{4}$$

where $\mathbf{D} = \{\mathbf{d}_j\}_{j=1}^J$ is the 3-D convolutional dictionary and $\mathbf{H} = \{\mathbf{h}_j\}_{j=1}^J$ denotes the corresponding sparse coefficients. Here, we use the shared 3-D filters (dictionaries) for all the bands to extract the local spatial–spectral variations.

Given (3) and (4), our goal is to jointly estimate $\mathbf{C}$ and $\mathbf{U}$ by exploiting their respective sparsity priors. The overall optimization problem is formulated as follows:

$$\min_{\mathbf{S},\mathbf{H}} \frac{1}{2} \|\mathbf{Y} - \mathbf{K} \otimes \mathbf{S} - \mathbf{D} \star \mathbf{H}\|_F^2 + \lambda_1 \|\mathbf{S}\|_1 + \lambda_2 \|\mathbf{H}\|_1 \tag{5}$$

where $\mathbf{K} = \text{concat}(\mathbf{K}_1, \ldots, \mathbf{K}_B)$, $\otimes$ denotes the depthwise convolution across spectral bands and λ_1 and λ_2 are the regularization parameters controlling the sparsity level. $\|\cdot\|_1$ norm refers to the L_1 norm, which computes the sum of the absolute values of the elements. This norm is widely used to promote sparsity in optimization problems, encouraging many coefficients to become 0.

In addition to global and local structures, HSIs exhibit nonlocal spatial self-similarity, arising from repetitive spatial patterns across the image. To effectively leverage this characteristic, we incorporate a data-driven regularization term $\mathcal{R}(\mathbf{S})$. Furthermore, the local spatial–spectral correlation primarily captures fine-grained image details, which are crucial for accurate reconstruction. To better exploit these details, we introduce another data-driven regularization term $\mathcal{R}(\mathbf{H})$. The resulting objective function is therefore formulated as follows:

$$\begin{aligned}\min_{\mathbf{S},\mathbf{H}} \frac{1}{2} \|\mathbf{Y} - \mathbf{K} \otimes \mathbf{S} - \mathbf{D} \star \mathbf{H}\|_F^2 + \lambda_1 \|\mathbf{S}\|_1 \\ + \lambda_2 \|\mathbf{H}\|_1 + \mu_1 \mathcal{R}(\mathbf{S}) + \mu_2 \mathcal{R}(\mathbf{H})\end{aligned} \tag{6}$$

where μ_1 and μ_2 balance the contributions of the data-driven regularizations.

To solve (6), we adopt an alternating optimization strategy that updates $\mathbf{S}$ and $\mathbf{H}$ iteratively

$$\min_{\mathbf{S}} \frac{1}{2} \|\mathbf{Y} - \mathbf{K} \otimes \mathbf{S} - \mathbf{D} \star \mathbf{H}\|_F^2 + \lambda_1 \|\mathbf{S}\|_1 + \mu_1 \mathcal{R}(\mathbf{S}) \tag{7}$$

$$\min_{\mathbf{H}} \frac{1}{2} \|\mathbf{Y} - \mathbf{K} \otimes \mathbf{S} - \mathbf{D} \star \mathbf{H}\|_F^2 + \lambda_2 \|\mathbf{H}\|_1 + \mu_2 \mathcal{R}(\mathbf{H}). \tag{8}$$

Each subproblem is a constrained sparse coding task and is solved via proximal gradient descent. The iterative updates at the $(t+1)$th step are given by

$$\mathbf{H}^{(t+1)} = \text{Net}_2\left(\text{Soft}_{\theta_2}\left(\mathbf{H}^{(t)} + \mathbf{D}^{\mathrm{T}} \star \left(\mathbf{Y} - \mathbf{K} \otimes \mathbf{S}^{(t+1)} - \mathbf{D} \star \mathbf{H}^{(t)}\right)\right)\right) \tag{9}$$

where $(\cdot)^{\mathrm{T}}$ denotes the transposed convolution and $\text{Soft}_\theta(\cdot)$ is the soft-thresholding function

$$\text{Soft}_\theta(x) = \text{sign}(x) \cdot \max(|x| - \theta, 0). \tag{10}$$

The modules Net_1 and Net_2 serve as learned regularizers, corresponding to the constraints $\mathcal{R}(\mathbf{S})$ and $\mathcal{R}(\mathbf{H})$, considering that the neural network can serve as a proximal operator. Once $\mathbf{S}$ and $\mathbf{H}$ are inferred, the clean HSI is reconstructed as follows:

$$\widehat{\mathbf{X}} = \mathbf{K} \otimes \mathbf{S} + \mathbf{D} \star \mathbf{H}. \tag{11}$$

B. DECSC Layer

1) Weight-Tied CSC Layers: In fact, both update steps in (9) share a common structural form that seeks the fixed-point solution of α

$$\alpha^{(t+1)} = \text{Net}\left(\text{Soft}_\theta\left(\alpha^{(t)} + \mathbf{E}^{\mathrm{T}} \star \left(\mathbf{z} - \mathbf{E} \star \alpha^{(t)}\right)\right)\right) \tag{12}$$

where $\mathbf{z}$ represents the noisy image injected into each layer. By introducing a set of parameters $\Theta = \{\theta, \mathbf{W}_E, \mathbf{E}\}$ with $\mathbf{W}_E = \mathbf{E}^{\mathrm{T}}$, (12) can be transformed into a learnable network architecture $\alpha^{(t+1)} = f_\Theta(\alpha^{(t)}, \mathbf{z})$ defined as follows:

$$f_\Theta\left(\alpha^{(t)}, \mathbf{z}\right) = \text{Net}\left(\text{Soft}_\theta\left(\alpha^{(t)} + \mathbf{W}_E \star \left(\mathbf{z} - \mathbf{E} \star \alpha^{(t)}\right)\right)\right). \tag{13}$$

Equation (13) is a weight-tied architecture with residual connections from the input to each layer until convergence. In other words, each layer incrementally refines the output by building upon the updates from the previous iteration. As the depth increases, the magnitude of these updates gradually diminishes, leading to a regime of diminishing returns, where additional layers contribute progressively less until the network reaches a stable equilibrium. As pointed in [15], such a design offers several advantages. First, weight sharing serves as a form of implicit regularization that stabilizes training and promotes generalization. Second, it significantly reduces the number of trainable parameters, resulting in more compact models. Third, unlike deep unfolding models that require a predefined, fixed number of layers, this iterative structure can theoretically be unrolled to arbitrary depth, aligning naturally with the principles of fixed-point optimization.

2) Forward and Backward Passes: Fig. 1 illustrates the processing flow of the model under the DEQ framework. As aforementioned, the output of a DEQ is the equilibrium point α^* where the predefined condition is met

$$\alpha^* = f_\Theta(\alpha^*, \mathbf{z}). \tag{14}$$

A naive way is to obtain α^* by iteratively running (12) till convergence, which is time-consuming. Instead, the equilibrium point can be faster obtained with any black-box root-finding algorithm. In our implementation, we use the Anderson acceleration procedure, which uses the past updates to identify promising directions to move during the current update to find the equilibrium point

$$\alpha^{(t+1)} = (1-\beta) \sum_{i=0}^{m} \gamma_i^{(t)} \alpha^{(t-i)} + \beta \sum_{i=0}^{m} \gamma_i^{(t)} f_\Theta\left(\alpha^{(t-i)}, \mathbf{z}\right) \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

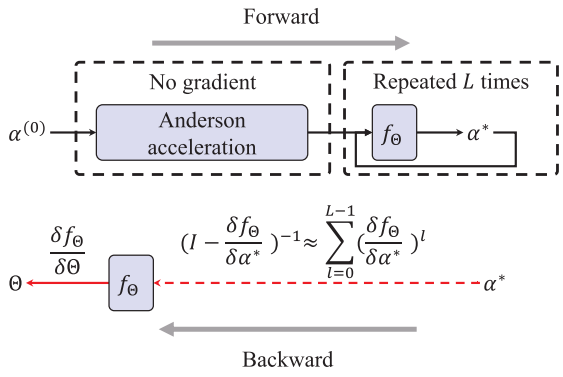


Fig. 1. DEQ-based inference of the proposed model. Forward pass uses Anderson acceleration; backward pass uses an unrolling-based phantom gradient.

for the mixing parameter $\beta > 0$. Defining the residual $g_\Theta(\alpha^{(t)}, \mathbf{z}) = f_\Theta(\alpha^{(t)}, \mathbf{z}) - \alpha^{(t)}$, γ is calculated as follows:

$$\arg\min_{\gamma} \|\mathbf{G}\gamma\|_2^2, \quad \text{s.t.} \sum_{i=0}^{m} \gamma_i = 1 \tag{16}$$

where $\mathbf{G} = [g_\Theta(\alpha^{(t)}, \mathbf{z}), \ldots, g_\Theta(\alpha^{(t-m+1)}, \mathbf{z})]$ is a matrix containing the m past residuals.

Since the forward pass of DEQ does not rely on explicit iterations, gradient computation cannot be automatically performed with Pytorch or Tensorflow, during the backward propagation process. According to the derivation by Bai et al. [15], the gradient with respect to the network parameters is learned with constant memory using implicit differentiation and is calculated as follows:

$$\frac{\partial \alpha^*}{\partial \Theta} = \left(\mathbf{I} - \frac{\partial f_\Theta(\alpha^*, \mathbf{z})}{\partial \alpha^*}\right)^{-1} \frac{\partial f_\Theta(\alpha^*, \mathbf{z})}{\partial \Theta}. \tag{17}$$

Since the exact computation of the Jacobian-inverse $(\mathbf{I} - (\partial f_\Theta(\alpha^*, \mathbf{z})/\partial \alpha^*))^{-1}$ requires high computational costs, we approximates it using phantom gradient [53]. Specifically, consider the Neumann series expansion of the Jacobian inverse

$$\mathbf{I} + \frac{\partial f_\Theta}{\partial \alpha^*} + \left(\frac{\partial f_\Theta}{\partial \alpha^*}\right)^2 + \left(\frac{\partial f_\Theta}{\partial \alpha^*}\right)^3 \cdots \tag{18}$$

The gradient approximated using an L-term Neumann series is equivalent to the differentiation of unrolling $\alpha^* = f_\Theta(\alpha^*, \mathbf{z})$ for L steps

$$\frac{\partial \alpha^*}{\partial \Theta} = \sum_{l=0}^{L-1} \left(\frac{\partial f_\Theta}{\partial \alpha^*}\right)^l \frac{\partial f_\Theta}{\partial \Theta} \approx \left(\mathbf{I} - \frac{\partial f_\Theta}{\partial \alpha^*}\right)^{-1} \frac{\partial f_\Theta}{\partial \Theta}. \tag{19}$$

As shown in Fig. 1, we therby adopt the unrolling-based phantom gradient to approximate the gradient to obtain an exact solution.

C. Network Implementation

1) Overall Architecture: Fig. 2 illustrates the architecture and implementation details of the network layer, showing how the components of $\mathbf{S}^{(t)}$ and $\mathbf{H}^{(t)}$ are updated in the proposed framework. With the DEQ CSC layers introduced in Section III-B, the iterative updates in (9) can be reformulated as two alternating layers, i.e., GIC layer and LSU layer

$$\begin{aligned}\mathbf{S}^{(t+1)} &= \text{Net}_1\left(\text{Soft}_{\theta_1}\left(\mathbf{S}^{(t)} + \mathbf{W}_K \otimes \left(\mathbf{Y} - \mathbf{K} \otimes \mathbf{S}^{(t)} - \mathbf{D} \star \mathbf{H}^{(t)}\right)\right)\right)\\ \mathbf{H}^{(t+1)} &= \text{Net}_2\left(\text{Soft}_{\theta_2}\left(\mathbf{H}^{(t)} + \mathbf{W}_D \star \left(\mathbf{Y} - \mathbf{K} \otimes \mathbf{S}^{(t+1)} - \mathbf{D} \star \mathbf{H}^{(t)}\right)\right)\right)\end{aligned} \tag{20}$$

where $\{\mathbf{W}_K, \mathbf{K}, \mathbf{W}_D, \mathbf{D}, \theta_1, \theta_2\}$ denote the set of learnable parameters, with the following relationships: $\mathbf{W}_K = \mathbf{K}^{\mathrm{T}}$ and $\mathbf{W}_D = \mathbf{D}^{\mathrm{T}}$.

Furthermore, we explicitly define an image reconstruction layer connected to the GIC and LSU layers. This layer takes the equilibrium representations $\mathbf{S}^*$ and $\mathbf{H}^*$ as inputs and reconstructs the clean HSI as follows:

$$\widehat{\mathbf{X}} = \mathbf{K} \otimes \mathbf{S}^* + \mathbf{D} \star \mathbf{H}^*. \tag{21}$$

2) Swin Transformer Module: Since $\mathbf{S}^{(t)}$ captures the globally shared spatial structure across bands, we introduce multiple stacked Swin Transformer blocks within the GIC component to efficiently capture its nonlocal properties, corresponding to the regularization term $\mathcal{R}(\mathbf{S})$ in (6). The Swin Transformer is adopted, considering its computational efficiency. Specifically, we divide $\mathbf{S}^{(t)}$ into multiple windows. For the tokens $\mathbf{P}$ within a window, the query, key, and value of the ith head are computed by linear projections using parameters $\mathbf{W}_i^q$, $\mathbf{W}_i^k$, and $\mathbf{W}_i^v$, respectively. Afterward, the attention output for the ith head is computed as follows:

$$\text{head}_i = \text{Softmax}\left(\frac{\left(\mathbf{W}_i^q \mathbf{P}\right)\left(\mathbf{W}_i^k \mathbf{P}\right)^{\mathrm{T}}}{\sqrt{d_i}}\right)\mathbf{W}_i^v \mathbf{P} \tag{22}$$

where d_i represents the feature dimension of the ith head. The outputs from all heads are concatenated and further projected to obtain the final result

$$\text{MSA}(\mathbf{P}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\,\mathbf{W}^o \tag{23}$$

where h is the number of heads and W^o is the projection matrix.

3) Detail Enhancement Module: The detail enhancement module, composed of a difference convolution and a spatial attention, is designed to strengthen the preservation of fine image details in the LSU component and corresponds regularization term $\mathcal{R}(\mathbf{H})$ in (6). It is formulated as follows:

$$\begin{aligned}\mathbf{S} &= \mathbf{S} + \text{DConv}(\mathbf{S})\\ \mathbf{S} &= \mathbf{S} + \mathbf{S} \bigodot \text{Conv}_1(\text{Conv}_1(\mathbf{S}))\end{aligned} \tag{24}$$

where Conv_1 is the $3 \times 3 \times 3$ 3-D convolution and DConv denotes the difference convolutional operator [56]. Due to the fact that conventional CNNs are optimized from random initialization, they often struggle to effectively focus on extracting image edges and fine details during training. In contrast, traditional edge detection operators characterize abrupt intensity changes and fine structural features by leveraging differential information from the image. Therefore, DConv [56] combines conventional edge detection operators with CNNs, enabling more precise capture of image gradient information while fully exploiting the powerful representational capability of deep learning. As illustrated in Fig. 3, we extend conventional edge detection operators to design a DConv module

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

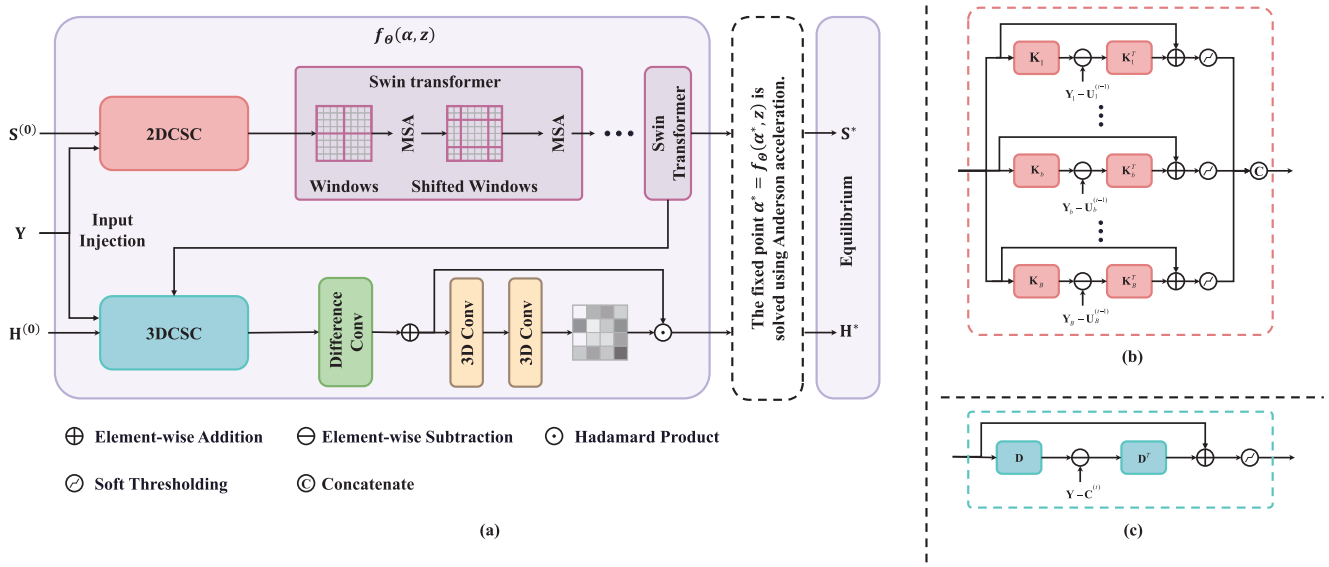


Fig. 2. Illustration of (a) DECSC architecture. The network leverages a DEQ model to transform the physical model, which incorporates (b) 2-D CSC for GIC structure learning and (c) 3-D CSC for LSU structure learning, into a fully learnable architecture.

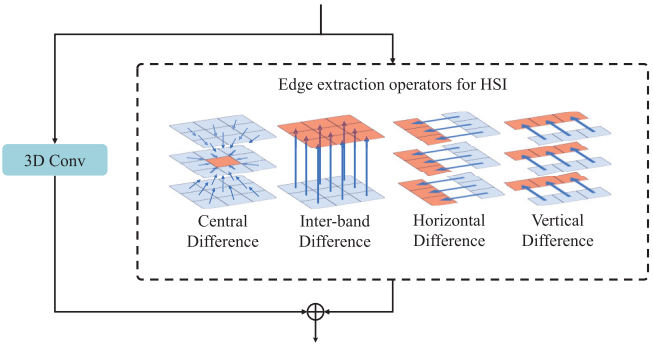


Fig. 3. Difference convolutional module consists of a 3-D convolution and four HSI-specific edge extraction operators.

tailored for HSI. It consists of a standard 3-D convolution for extracting intensity information, along with four edge extraction operators specifically designed for HSI to capture gradient information. In particular, the central difference operator enhances image sharpness, the interband difference operator captures variations along the spectral dimension, and the horizontal difference and vertical difference operators in the spatial domain extract horizontal and vertical edge information, respectively. This intermediate output is then passed through two cascaded 3-D convolutional layers, which serve as an attention that adaptively emphasizes regions rich in image detail information.

4) Training Loss: The loss function of our DEQCSC is the Euclidean distance between the estimated clean HSI and the ground truth

$$\mathcal{L}_\Theta = \frac{1}{N}\sum_{i}^{N}\left\|\widehat{\mathbf{X}}_i - \mathbf{X}_i\right\|_F^2 \tag{25}$$

where Θ represents the network parameters, N is the total number of training samples, and $\|\cdot\|_F^2$ denotes the Frobenius norm.

IV. Experiment

In this section, we first present the experiment settings and implementation details, followed by an analysis of the results for synthetic noise and real-world noise removal experiments. Finally, we present a series of ablation studies.

A. Experimental Settings and Implementation Details

1) Benchmarked Models: The comparison involves 12 methods, comprising seven model-driven methods, three data-driven methods, and two hybrid-driven methods. The model-driven methods include BM4D [57], MTSNMF [58], LLRT [59], NGMeet [21], LRMR [60], E-3DTV [28], and 3DlogTNN [22]. The data-driven methods include SST [32], TRQ3D [61], and SERT [33], while the hybrid-driven methods based on sparse priors include T3SC [44] and MTSNN++ [47].

2) Metrics: To quantitatively evaluate the performance of the models, we adopt standard metrics including peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and spectral angle mapper (SAM), where higher PSNR and SSIM values, along with lower SAM values, reflect better denoising performance.

3) Noise Patterns: Nonindependent and identically distributed (non-i.i.d.) Gaussian noise and mixture noise were considered in the synthetic experiment. In addition, we adopted the "noise with spectrally correlated variance" model proposed by Bodrito et al. [44]. The detailed noise settings are described as follows.

1) *Non-i.i.d. Gaussian Noise:* Each band is contaminated with Gaussian noise, where the standard deviation σ is uniformly sampled from fixed intervals, i.e., [0, 15], [0, 55], and [0, 95].
2) *Mixture Noise:* In addition to being contaminated by non-i.i.d. distributed Gaussian noise with intensity [0, 95], every one-third of the bands is further degraded by the additional noise types: impulse noise with intensities

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON OF DIFFERENT METHODS ON 50 TESTING HSIS FROM ICVL DATASET. THE TOP THREE VALUES ARE MARKED AS RED, BLUE, AND GREEN

σ	Index	Noisy	Model-driven							Data-driven			Hybrid-driven		
			BM4D [57]	MTSNMF [58]	LLRT [59]	NGMeet [21]	LRMR [60]	E-3DTV [28]	3DlogTNN [22]	SST [32]	TRQ3D [61]	SERT [33]	T3SC [44]	MTSNN++ [47]	DECSC (Ours)
[0,15]	PSNR↑	33.18	44.39	45.39	45.74	39.63	41.50	46.05	43.89	**50.87**	46.43	**50.17**	49.68	48.86	**50.63**
	SSIM↑	.6168	.9683	.9592	.9657	.8612	.9356	.9811	.9902	**.9938**	.9878	**.9976**	.9912	.9917	**.9936**
	SAM↓	.3368	.0692	.0845	.0832	.2144	.1289	.0560	**.0150**	.0298	.0437	**.0277**	.0486	.0346	**.0265**
[0,55]	PSNR↑	21.72	37.63	38.02	36.80	31.53	31.50	40.20	33.37	**46.39**	44.64	**46.33**	45.15	43.88	**46.92**
	SSIM↑	.2339	.9008	.8586	.8285	.6785	.6233	.9505	.6892	**.9872**	.9840	**.9950**	.9810	.9794	**.9876**
	SAM↓	.7012	.1397	.2340	.2316	.4787	.3583	.0993	.2766	**.0457**	.0487	**.0372**	.0652	.0528	**.0348**
[0,95]	PSNR↑	17.43	34.71	34.81	31.89	27.62	27.00	37.80	24.53	**44.83**	43.54	**44.47**	43.10	42.15	**45.64**
	SSIM↑	.1540	.8402	.7997	.6885	.5363	.4208	.9279	.4251	**.9838**	.9806	**.9929**	.9734	.9720	**.9848**
	SAM↓	.8893	.1906	.3266	.3444	.6420	.5142	.1317	.6087	**.0513**	.0523	**.0446**	.0747	.0665	**.0387**
Mixture	PSNR↑	13.21	23.36	27.55	18.23	23.61	23.10	34.90	17.52	**39.22**	**39.73**	39.13	34.09	38.90	**42.67**
	SSIM↑	.0841	.4275	.6743	.1731	.4448	.3463	.9041	.2389	**.9626**	.9491	**.9678**	.9052	.9531	**.9756**
	SAM↓	.9124	.5476	.5326	.6873	.6252	.5144	.1468	.6905	**.0743**	**.0869**	.0963	.2340	.0885	**.0625**
Corr	PSNR↑	28.22	41.15	42.44	41.92	35.82	39.32	43.58	41.49	**47.59**	46.26	**48.66**	47.33	46.83	**48.06**
	SSIM↑	.4640	.8963	.9221	.9080	.7891	.9081	.9733	.9709	**.9904**	.9870	**.9966**	.9858	.9871	**.9891**
	SAM↓	.4601	.1582	.1121	.1547	.3113	.1212	.0601	.0574	**.0258**	.0403	**.0351**	.0524	.0399	**.0298**

ranging from 0.1 to 0.7, strip noise affecting 5%–15% of columns, and deadline noise.

3) *Noise With Spectrally Correlated Variance:* Each band is contaminated with Gaussian noise whose standard deviation σ varies continuously across bands following a Gaussian distribution. Specifically, for each band $i \in [0, B-1]$, σ is defined as follows:

$$\sigma_i = \beta \exp\left[-\frac{1}{4\eta^2}\left(\frac{i}{c}-\frac{1}{2}\right)^2\right] \tag{26}$$

where $\beta = 23.08$ and $\eta = 0.157$ as in [44].

4) Implementation Details: The proposed DECSC model was implemented in PyTorch and trained using the Adam optimizer, starting with an initial learning rate of 0.0001 and a batch size of 8. The learning rate was reduced by half every 10 epochs, and training was conducted for 30 epochs on a Linux machine equipped with an Intel[1] Xeon[1] E5-2650 v4 GHz and four NVIDIA GeForce RTX 4090 GPUs. For the GIC component, the dictionary size was set to 192 with a convolutional filter size of 9×9, and the Swin Transformer consists of four stacked stages with a window size of 4×4. For the LSU component, the dictionary size was set to 96, also with a convolutional filter size of $9 \times 9 \times 3$. The Anderson acceleration was employed to efficiently compute the equilibrium point (fixed point), while a rolling-based phantom gradient strategy with a truncation length of $L = 5$ was used to approximate gradients during backpropagation. For synthetic noise removal, all models were pretrained on the ICVL dataset and subsequently fine-tuned on the corresponding test datasets. In real noise scenarios, where ground truth is unavailable, band splitting was applied directly during inference.

B. Synthetic Noise Removal

To comprehensively evaluate the DECSC's ability to remove synthetic noise, we conducted denoising experiments on both close-range HSIs, i.e., the ICVL dataset, and remote-range HSI, i.e., the Houston 2018 HSI, using the synthetic noise patterns described earlier.

1) ICVL Dataset: The ICVL dataset spans a spectral range of 400–700 nm. For testing, each HSI is cropped to a size of $512 \times 512 \times 31$. Table I presents a quantitative comparison of the denoising performance. As data-driven neural network methods learn mappings from noisy to clean HSIs using large-scale data, they are better suited to capture the intrinsic structures of HSIs than methods relying on hand-crafted priors, resulting in a notable performance advantage. Furthermore, the performance of model-driven approaches that depend on precise noise intensity estimation may degrade when noise levels vary independently across spectral bands. All data-driven methods evaluated in this study utilize transformer-based architectures. Among them, SST and SERT exhibit strong performance due to the incorporation of spectral attention mechanisms and low-rank memory units, respectively. The comparative hybrid-driven methods evaluated here are based on sparse priors. Although MTSNN++ shares a similar design philosophy with DECSC, it does not adequately account for nonlocal self-similarities. In contrast, the proposed DECSC outperforms others by jointly modeling global inter-band structural consistency, local spatial–spectral correlations, and nonlocal self-similarities. Moreover, the DEQ framework further enhances the denoising performance compared to the deep unfolding-based MTSNN++. However, we also observe that our DECSC lags behind SERT in the Corr noise pattern, possibly due to SERT's superior ability to capture global spectral correlations through its low-rank memory unit.

Fig. 4 presents a visual comparison of different methods. To better highlight the denoising performance, we selected three spectral bands with severe noise corruption. Consistent with the quantitative results, data-driven and hybrid-driven methods generally outperform purely model-driven approaches. Model-driven methods often leave residual noise, color distortions, and blurred details, while data-driven and hybrid-driven methods achieve more effective noise suppression. Nonetheless, T3SC, TRQ3D, and MTSNN++ still exhibit blur patterns, whereas SST and SERT produce sharper images but suffer

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

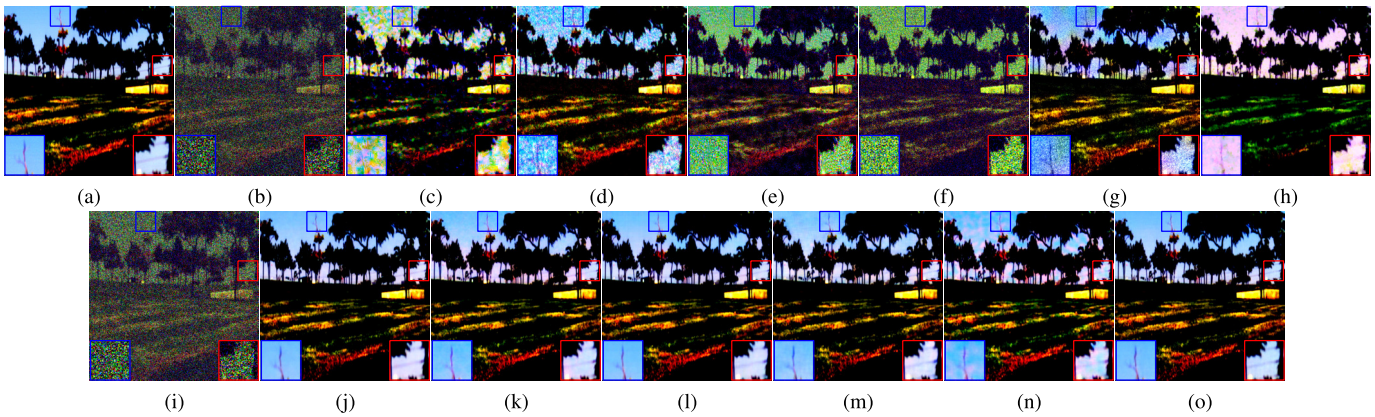


Fig. 4. Denoising results on the Nachal_0823 – 1214 HSI from the ICVL dataset under the non-i.i.d. Gaussian noise with $\sigma \in [0, 95]$. The false-color images are generated by combining bands 31, 17, and 2. (a) Clean. (b) Noisy. (c) BM4D [57]. (d) MTSNMF [58]. (e) LLRT [59]. (f) NGMeet [21]. (g) LRMR [60]. (h) E-3DTV [28]. (i) 3DlogTNN [22]. (j) SST [32]. (k) TRQ3D [61]. (l) SERT [33]. (m) T3SC [44]. (n) MTSNN++ [47]. (o) DECSC.

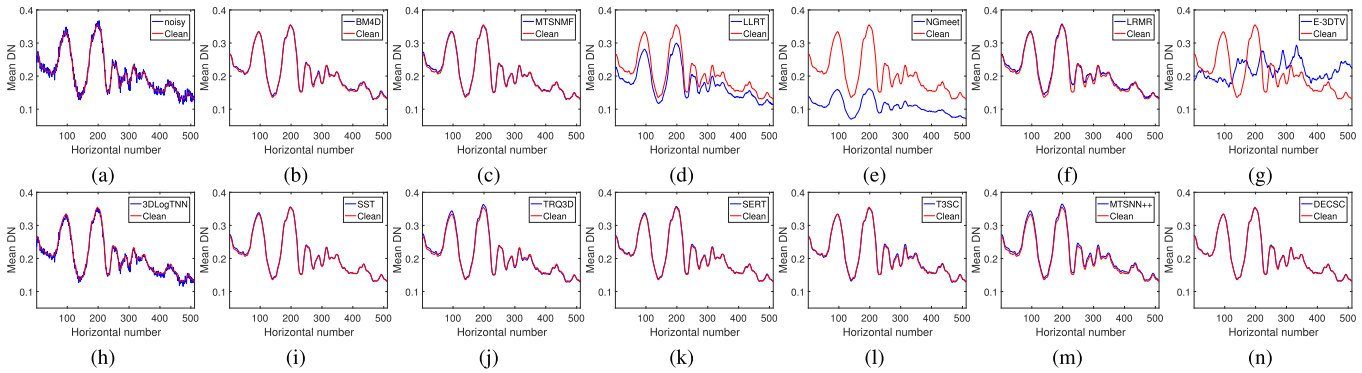


Fig. 5. Row mean profiles of band 28 for the Nachal_0823 – 1214 HSI from the ICVL dataset under the non-i.i.d. Gaussian noise with $\sigma \in [0, 95]$. (a) Noisy. (b) BM4D [57]. (c) MTSNMF [58]. (d) LLRT [59]. (e) NGMeet [21]. (f) LRMR [60]. (g) E-3DTV [28]. (h) 3DlogTNN [22]. (i) SST [32]. (j) TRQ3D [61]. (k) SERT [33]. (l) T3SC [44]. (m) MTSNN++ [47]. (n) DECSC.

from slight color shifts. In contrast, the proposed method demonstrates superior preservation of both image detail and color fidelity, benefiting from the guidance of low-noise bands and the effectiveness of the detail enhancement module.

Fig. 5 shows the comparison of the row mean profiles of the restoration results obtained by different methods. Except for some model-driven approaches, most methods are able to recover results that are globally similar to the clean HSI, though local deviations still exist. It is worth noting that the row mean profile, which represents the global trend along the vertical spatial dimension by averaging, inherently suppresses local spatial variations. As a result, it can appear close to the ground truth even when spatial details are degraded, leading to discrepancies with pixel-level metrics such as PSNR and with visual quality, both of which are sensitive to fine spatial distortions. Methods with strong global regularization often maintain accurate row mean profiles but may suffer from excessive spatial smoothing or insufficient noise removal, thus lowering PSNR and perceived quality. In contrast, the proposed DECSC achieves results that are noticeably closer to the clean HSI in both global trends and fine details, demonstrating its strong denoising ability.

2) Houston 2018 HSI: The Houston 2018 HSI contains 1202×4172 pixels and spans 48 spectral bands ranging from 380 to 1050 nm. For testing, we selected the last 46 relatively clean bands and cropped a 512×512 patch from the central region as the testing sample. The remaining regions were divided into 64×64 patches for model fine-tuning. Table II presents a quantitative comparison of denoising performance on the Houston 2018 HSI under various noise patterns. The results are largely consistent with those reported in Table I. Despite differences in imaging conditions across datasets, fine-tuning enables both data-driven and hybrid-driven methods to outperform purely model-driven approaches. The proposed DECSC achieves the best performance under non-i.i.d. Gaussian and mixture noise patterns, which we attribute to the robustness of its convolutional sparse dictionary specifically designed to capture GIC and LSU structures. Moreover, the DEQ model provides an infinite-depth network that inherits the robustness of the physical model while guaranteeing convergence.

Fig. 6 shows the comparison of the restoration results of different methods on the Houston 2018 HSI. As a remote sensing dataset, the Houston HSI is typically affected by mixture noise; thus, Fig. 6 focuses on restoration performance under the mixture noise pattern. Most model-driven methods, which are primarily designed for Gaussian noise, struggle with complex noise types, leading to lower visual quality. In contrast, data-driven and hybrid-driven methods generally yield better visual results due to the strong representational capacity of DNNs. Nevertheless, color distortions and detail over-smoothing remain common issues in many of these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

TABLE II

COMPARISON OF DIFFERENT METHODS ON HOUSTON 2018 HSI. THE TOP THREE VALUES ARE MARKED AS RED, BLUE, AND GREEN

σ	Index	Noisy	Model-driven							Data-driven			Hybrid-driven		
			BM4D [57]	MTSNMF [58]	LLRT [59]	NGMeet [21]	LRMR [60]	E-3DTV [28]	3DlogTNN [22]	SST [32]	TRQ3D [61]	SERT [33]	T3SC [44]	MTSNN++ [47]	DECSC (Ours)
[0,15]	PSNR↑	32.47	40.70	42.83	40.65	38.31	39.90	43.39	43.66	**50.53**	44.75	**49.60**	49.09	48.11	**51.35**
	SSIM↑	.6738	.9667	.9775	.9468	.8906	.9601	.9838	.9869	**.9959**	.9867	**.9981**	.9947	.9927	**.9961**
	SAM↓	.2562	.0550	.0357	.0712	.1279	.0612	.0427	.0327	**.0180**	.0284	**.0200**	.0214	.0232	**.0173**
[0,55]	PSNR↑	24.66	35.04	35.94	30.15	29.86	30.93	39.23	31.44	**47.64**	44.76	**45.65**	44.20	44.98	**49.05**
	SSIM↑	.3874	.9002	.8919	.6390	.6918	.7624	.9629	.6480	**.9926**	.9874	**.9955**	.9872	.9862	**.9933**
	SAM↓	.6484	.1014	.1387	.3295	.3707	.1688	.0690	.4463	**.0227**	.0307	**.0263**	.0316	.0292	**.0209**
[0,95]	PSNR↑	16.85	31.23	32.73	22.93	26.06	26.43	34.92	23.10	**43.10**	41.74	**42.10**	40.08	41.07	**43.25**
	SSIM↑	.2072	.7766	.8160	.3724	.5514	.5662	.9254	.4339	**.9834**	.9779	**.9911**	.9699	.9719	**.9834**
	SAM↓	.9201	.1961	.2160	.5802	.5400	.3017	.1117	.8329	**.0310**	.0389	**.0356**	.0466	.0434	**.0305**
Mixture	PSNR↑	11.72	22.76	25.86	15.58	22.36	21.84	30.69	15.43	**35.85**	**36.23**	34.86	28.66	34.71	**38.65**
	SSIM↑	.0843	.4762	.6933	.1386	.5169	.3914	.8582	.1965	**.9449**	.9363	**.9643**	.8471	.9081	**.9581**
	SAM↓	.9778	.5168	.4977	.7652	.5728	.4857	.1316	.9033	**.0605**	**.0659**	.0842	.2280	.0817	**.0480**
Corr	PSNR↑	28.21	37.28	40.22	36.04	35.69	37.12	40.65	41.01	**46.54**	45.36	**48.32**	45.90	45.42	**46.32**
	SSIM↑	.5631	.8721	.9554	.8383	.8634	.9429	.9706	.9702	**.9910**	.9886	**.9973**	.9899	.9882	**.9904**
	SAM↓	.3649	.1466	.0525	.1597	.1834	.0761	.0493	.0423	**.0221**	.0268	**.0201**	.0251	.0265	**.0225**

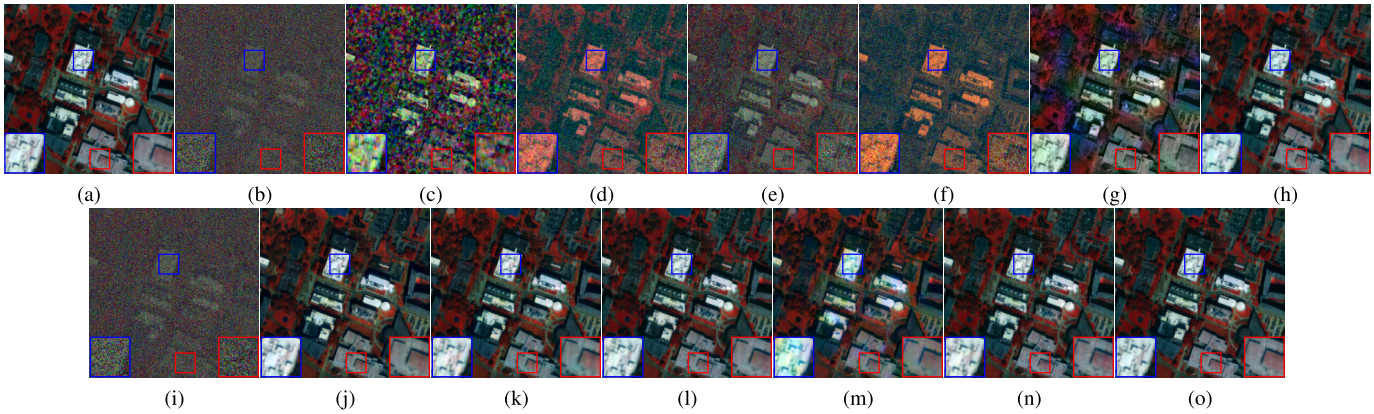


Fig. 6. Denoising results on the Houston 2018 HSI under the mixture noise. The false-color images are generated by combining bands 39, 20, and 4. (a) Clean. (b) Noisy. (c) BM4D [57]. (d) MTSNMF [58]. (e) LLRT [59]. (f) NGMeet [21]. (g) LRMR [60]. (h) E-3DTV [28]. (i) 3DlogTNN [22]. (j) SST [32]. (k) TRQ3D [61]. (l) SERT [33] (m) T3SC [44]. (n) MTSNN++ [47]. (o) DECSC.

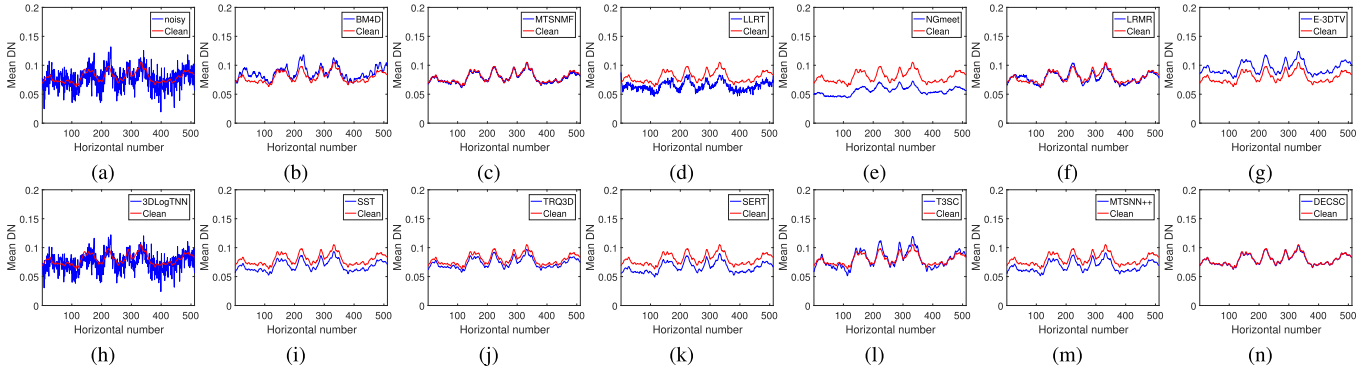


Fig. 7. Row mean profiles of band 39 for the Houston 2018 HSI under the mixture noise. (a) Noisy. (b) BM4D [57]. (c) MTSNMF [58]. (d) LLRT [59]. (e) NGMeet [21]. (f) LRMR [60]. (g) E-3DTV [28]. (h) 3DlogTNN [22]. (i) SST [32]. (j) TRQ3D [61]. (k) SERT [33]. (l) T3SC [44]. (m) MTSNN++ [47]. (n) DECSC.

methods. Among all compared approaches, the SERT and the proposed DECSC demonstrate visual results that are most consistent with the ground truth. Furthermore, Fig. 7 shows the comparison of the row mean profiles of the restoration results produced by different methods. The mixture noise poses a significant challenge for most methods, leading to evident overall shifts or residual noise in the restored results. Although MTSNMF yields reconstructions that are relatively closer to the clean HSI in terms of overall structure, noticeable local deviations still persist. In contrast, DECSC consistently produces profiles that are more aligned with the clean HSI.

C. Real-World Noise Removal

To further evaluate the denoising performance of DECSC in real-world scenarios, we selected two real-world remote sensing HSIs for experiments involving real-world noise removal.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

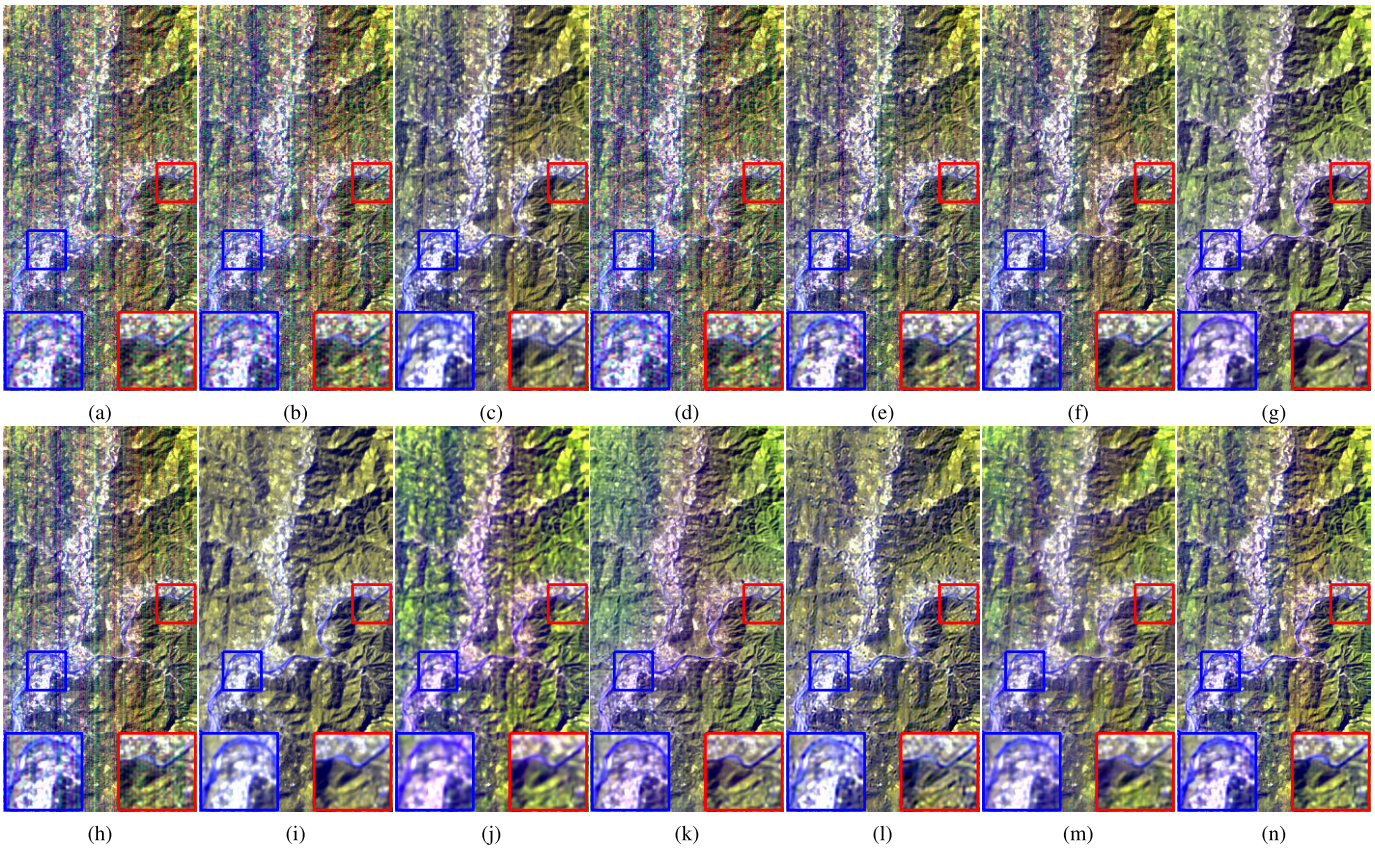


Fig. 8. Denoising results on the EO-1 HSI. The false-color images are generated by combining bands 97, 95, and 1. (a) Noisy. (b) BM4D [57]. (c) MTSNMF [58]. (d) LLRT [59]. (e) NGMeet [21]. (f) LRMR [60]. (g) E-3DTV [28]. (h) 3DlogTNN [22]. (i) SST [32]. (j) TRQ3D [61]. (k) SERT [33]. (l) T3SC [44]. (m) MTSNN++ [47]. (n) DECSC.

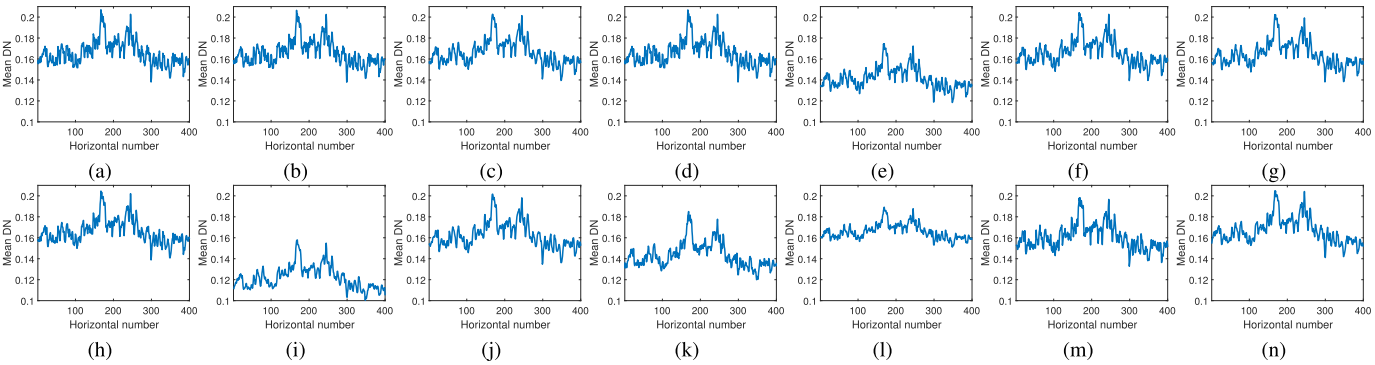


Fig. 9. Row mean profiles of band 159 for the EO-1 HSI. (a) Noisy. (b) BM4D [57]. (c) MTSNMF [58]. (d) LLRT [59]. (e) NGMeet [21]. (f) LRMR [60]. (g) E-3DTV [28]. (h) 3DlogTNN [22]. (i) SST [32]. (j) TRQ3D [61]. (k) SERT [33]. (l) T3SC [44]. (m) MTSNN++ [47]. (n) DECSC.

Due to the absence of corresponding ground truth, the evaluation was limited to a qualitative analysis of the restoration results.

1) Earth Observing-1 (EO-1) HSI: We selected an HSI captured by the EO-1 satellite, which covers a spectral range of 400–2500 nm. Following the experimental setup of Zhang et al. [60], a subimage of size $200 \times 400 \times 166$ was used for testing. Fig. 8 shows the comparison of the visual quality of restoration results produced by different methods. Most model-driven methods struggle to effectively remove stripe noise. E-3DTV, benefiting from its ability to capture correlations and differences across bands, produces relatively clear results even under severe noise. The differences between synthetic and real-world noise pose challenges for data-driven and hybrid-driven methods, leading to blurring in the results of methods like SST, TRQ3D, T3SC, and MTSNN++. In contrast, SERT and DECSC not only achieve effective denoising but also preserve well-defined edges and structural integrity in the reconstructions. The superior performance of SERT can be attributed to its rectangular self-attention mechanism, while DECSC's advantage lies in the difference convolution's ability to capture edge information, complemented by an attention mechanism that adaptively enhances critical features. Fig. 9 shows the row mean profiles of band 159 for the EO-1 HSI. Although clean ground truth is unavailable in this real-world scenario, the profiles still provide insight into the smoothness

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

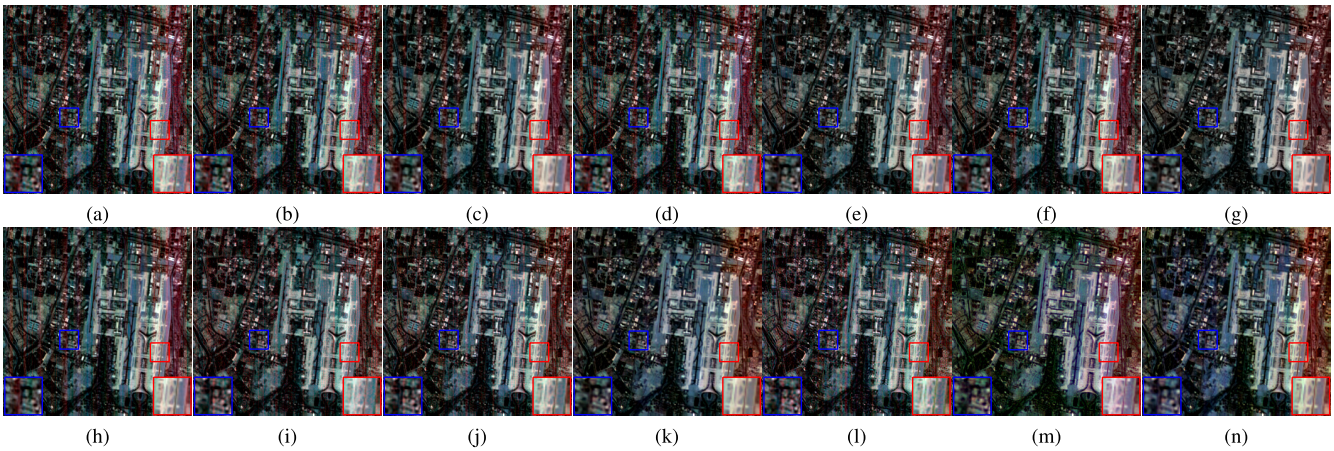


Fig. 10. Denoising results of CapitalAirport HSI collected from the GF-5 satellite. The false-color images are generated by combining bands 153, 107, and 94. (a) Noisy. (b) BM4D [57]. (c) MTSNMF [58]. (d) LLRT [59]. (e) NGMeet [21]. (f) LRMR [60]. (g) E-3DTV [28]. (h) 3DlogTNN [22]. (i) SST [32]. (j) TRQ3D [61]. (k) SERT [33]. (l) T3SC [44]. (m) MTSNN++ [47]. (n) DECSC.

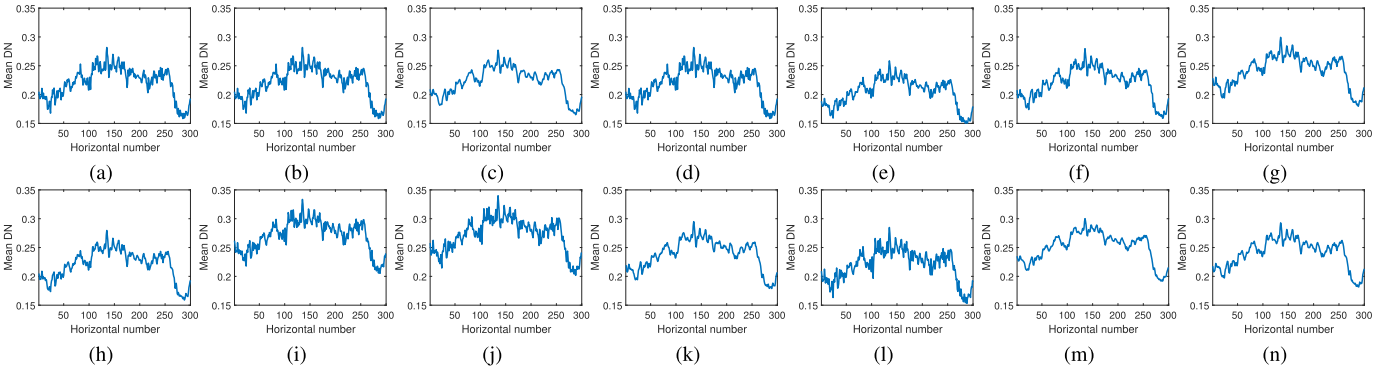


Fig. 11. Row mean profiles of band 154 for the CapitalAirport HSI collected from the GF-5 satellite. (a) Noisy. (b) BM4D [57]. (c) MTSNMF [58]. (d) LLRT [59]. (e) NGMeet [21]. (f) LRMR [60]. (g) E-3DTV [28]. (h) 3DlogTNN [22]. (i) SST [32]. (j) TRQ3D [61]. (k) SERT [33]. (l) T3SC [44]. (m) MTSNN++ [47]. (n) DECSC.

TABLE III
ABLATION STUDY ON THE CONTRIBUTION OF EACH MODULE

Swin	Detail Enhancement Module		PSNR↑	SSIM↑	SAM↓
	Difference Conv	Attention			
✓	✓	✓	45.64	.9848	.0387
✗	✓	✓	44.87	.9823	.0431
✓	✗	✓	45.32	.9839	.0410
✓	✓	✗	44.99	.9829	.0434

and consistency of the restored results. E-3DTV and DECSC achieve a favorable balance between smoothness and consistency with the value range of the noisy input, validating their robustness and effectiveness in practical scenarios.

2) Gaofen-5 (GF-5) CapitalAirport HSI: We selected an HSI captured by the GF-5 satellite, covering a spectral range of 400–2500 nm, for real-world noise removal experiments. In this study, a subimage of size $300\times300\times166$ was extracted for testing. As shown in Figs. 10 and 11, MTSNMF, SERT, and DECSC yield results whose value ranges are more consistent with the noisy input.

D. Network Analysis

In this section, we conduct a further analysis of the DEQCSC components and hyperparameter settings. All experiments are performed on the ICVL dataset under the non-i.i.d. Gaussian noise setting with noise levels in the range [0, 95].

1) Ablation Study: We conduct ablation studies on key components of our network to validate the effectiveness of the proposed design choices. Specifically, we evaluate the contributions of the Swin Transformer and the detail enhancement module by independently removing the Swin Transformer, the difference convolution, and the attention mechanism. Each variant was trained independently under identical settings. As shown in Table III, all three components significantly contribute to the model's overall performance. The results indicate that the Swin Transformer effectively reinforces nonlocal spatial self-similarities across bands, thereby enhancing denoising performance. The difference convolution preserves fine image details, improving the model's denoising capacity. Meanwhile, the attention mechanism guides the network to focus on important regions, further boosting denoising effectiveness.

2) Convergence Property: To further validate the relationship between our network and numerical optimization, we plot the changes in PSNR with respect to the number of layers during the forward process. As shown in Fig. 12, the PSNR value gradually reaches an optimal level and then stabilizes, with minimal changes as the number of layers increases. This phenomenon clearly demonstrates the promising relationship

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF MODEL COMPLEXITY AND EFFICIENCY ACROSS ALL METHODS

Index	Model-driven							Data-driven			Hybrid-driven		
	BM4D	MTSNMF	LLRT	NGMeet	LRMR	E-3DTV	3DlogTNN	SST	TRQ3D	SERT	T3SC	MTSNN++	**DECSC (Swin)**
#Param (M)	-	-	-	-	-	-	-	4.14	0.67	1.91	0.83	1.95	4.29
Time (s)	198.05	40	1673.8	512.13	274.92	55.55	188.51	1.83	1.44	0.36	0.57	56.96	36.95
FLOPS	-	-	-	-	-	-	-	67.61G	66.74G	29.92G	365.24M	34.71G	56.39T

TABLE V
ABLATION STUDY ON SWIN TRANSFORMER

Methods	Swin	Mamba
PSNR	45.64	**45.7**
SSIM	0.9848	**0.9851**
SAM	**0.0387**	0.0391
#Param (M)	**4.29**	5.05
Time (s)	36.95	**29.71**

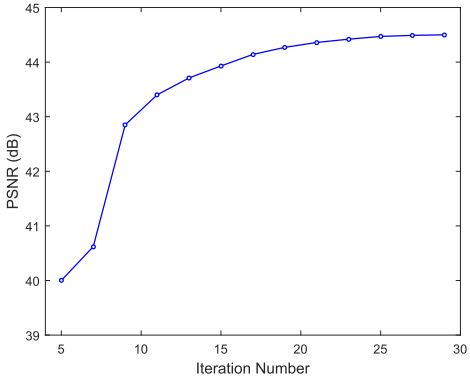


Fig. 12. As the number of iterations increases, the PSNR of the fixed-point solution improves steadily and asymptotically converges.

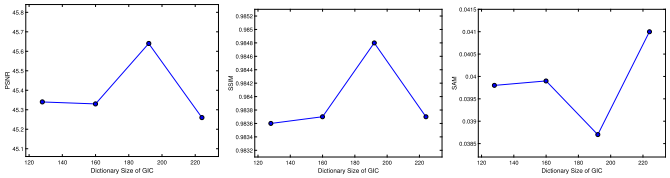


Fig. 13. Impact of the number of atoms in the GIC component on the denoising performance.

between the network and the physical model, which guarantees the convergence.

3) Impact of Dictionary Size: To evaluate the impact of the number of atoms on denoising performance, we conducted ablation experiments on both GIC and LSU components. Specifically, we first fixed the number of atoms in the LSU component to 96 and varied the number in the GIC component from 128 to 224 in increments of 32. As shown in Fig. 13, the denoising performance generally improved with an increasing dictionary size, with the best performance achieved at 192 atoms. Second, with the number of atoms in the GIC component fixed at 192, we varied the number of atoms in the LSU component from 32 to 128 in increments of 32. As shown in Fig. 14, the optimal performance was obtained when the number of atoms was set to 96.

4) Impact of Neumann Series: We conducted an ablation study on the Neumann series L to investigate its impact on denoising performance. As shown in Fig. 15, the overall denoising performance improves with increasing approximation accuracy, and the best performance is achieved when $L = 5$.

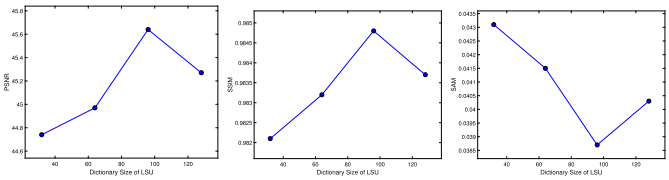


Fig. 14. Impact of the number of atoms in the LSU component on the denoising performance.

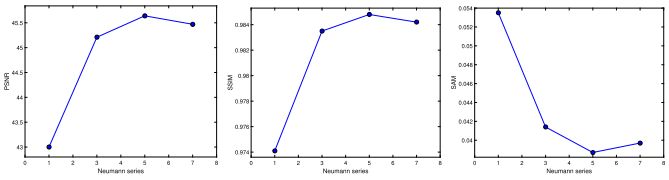


Fig. 15. Impact of the Neumann series parameter L on the denoising performance.

5) Comparison of Model Complexity and Efficiency: As shown in Table IV, our DECSC model maintains a comparable parameter count to the SST model while achieving significantly better performance. However, we acknowledge that our method lags behind in running time due to its iterative forward pass, which stops only upon reaching an equilibrium point. This process is inherently more time-consuming and requires more FLOPs than conventional neural networks with a fixed number of layers.

6) Replacing Transformer With Mamba: Given the substantial computational overhead of Transformers for long sequences, we further adopt Mamba to model global dependencies, leveraging its proven efficiency in HSI processing [37], [62], [63]. Specifically, we use the VMamba architecture [64], which performs directional scanning along four spatial orientations to capture global interactions within the GIC. Both Transformer and VMamba variants use an identical four-layer stacking configuration for a fair comparison. As shown in Table V, the Mamba-based model achieves comparable performance to the Transformer while reducing computational complexity and inference time.

V. CONCLUSION

This article introduces a novel DECSC framework for robust HSI denoising. By modeling the global shared spatial structure and local spatial–spectral structure, the denoising performance is significantly improved. In addition, the integration of transformer blocks and a detail enhancement module further boosts denoising by capturing nonlocal spatial self-similarities and local details. Thanks to the DEQ approach,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

the iterative optimization of the CSC model is effectively transformed into a learnable network that maintains physical interpretability and convergence guarantees. In future work, we plan to explore additional priors to better capture the unique structural properties of HSIs.

REFERENCES

[1] S. Jia, Y. Wang, S. Jiang, and R. He, "A center-masked transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510416.

[2] B. Fei, "Hyperspectral imaging in medical applications," in *Data Handling in Science and Technology*, vol. 32. Amsterdam, The Netherlands: Elsevier, 2019, pp. 523–565.

[3] K. Kersting et al., "Pre-symptomatic prediction of plant drought stress using Dirichlet-aggregation regression on hyperspectral images," in *Proc. AAAI Conf. Artif. Intell.*, vol. 26, 2021, pp. 302–308.

[4] F. Xiong, Y. Sun, J. Zhou, J. Lu, and Y. Qian, "Spatial–spectral–temporal correlation filter for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506613.

[5] Z. Li et al., "Multi-domain universal representation learning for hyperspectral object tracking," *Pattern Recognit.*, vol. 162, Jun. 2025, Art. no. 111389.

[6] Z. Li, F. Xiong, J. Zhou, J. Lu, Z. Zhao, and Y. Qian, "Material-guided multiview fusion network for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509415.

[7] S. Gu, W. Zuo, Q. Xie, D. Meng, X. Feng, and L. Zhang, "Convolutional sparse coding for image super-resolution," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1823–1831.

[8] P. Bao et al., "Convolutional sparse coding for compressed sensing CT reconstruction," *IEEE Trans. Med. Imag.*, vol. 38, no. 11, pp. 2607–2619, Nov. 2019.

[9] X. Hu, F. Heide, Q. Dai, and G. Wetzstein, "Convolutional sparse coding for RGB+NIR imaging," *IEEE Trans. Image Process.*, vol. 27, no. 4, pp. 1611–1625, Apr. 2018.

[10] F. Xiong, M. Ye, J. Zhou, and Y. Qian, "Spatial–spectral convolutional sparse neural network for hyperspectral image denoising," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1225–1228.

[11] H. Yin and H. Chen, "Deep content-dependent 3-D convolutional sparse coding for hyperspectral image denoising," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4125–4138, 2024.

[12] K. Tu, F. Xiong, G. Fu, and J. Lu, "Multitask hyperspectral image convolutional sparse coding-denoising network," *J. Image Graph.*, vol. 29, no. 1, pp. 280–292, 2024.

[13] D. Gilton, G. Ongie, and R. Willett, "Deep equilibrium architectures for inverse problems in imaging," *IEEE Trans. Comput. Imag.*, vol. 7, pp. 1123–1133, 2021.

[14] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing," *IEEE Signal Process. Mag.*, vol. 38, no. 2, pp. 18–44, Mar. 2021.

[15] S. Bai, J. Z. Kolter, and V. Koltun, "Deep equilibrium models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019.

[16] Y. Zhao, S. Zheng, and X. Yuan, "Deep equilibrium models for snapshot compressive imaging," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, vol. 37, no. 3, pp. 3642–3650.

[17] Z. Geng, A. Pokle, and J. Z. Kolter, "One-step diffusion distillation via deep equilibrium models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024, pp. 41914–41931.

[18] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534114.

[19] J. Ye, F. Xiong, J. Zhou, and Y. Qian, "Iterative low-rank network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528015.

[20] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, Y. Chen, and W. He, "Double-factor-regularized low-rank tensor factorization for mixed noise removal in hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8450–8464, Dec. 2020.

[21] W. He et al., "Non-local meets global: An iterative paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

[22] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, T.-X. Jiang, T.-H. Ma, and T.-Y. Ji, "Mixed noise removal in hyperspectral image via low-fibered-rank regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 734–749, Jan. 2020.

[23] T. Xie, S. Li, and J. Lai, "Adaptive rank and structured sparsity corrections for hyperspectral image restoration," *IEEE Trans. Cybern.*, vol. 52, no. 9, pp. 8729–8740, Sep. 2022.

[24] Y. Chen, W. Cao, L. Pang, and X. Cao, "Hyperspectral image denoising with weighted nonlocal low-rank model and adaptive total variation regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544115.

[25] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, and B. Zhang, "Decomposable nonlocal tensor dictionary learning for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2949–2956.

[26] Y.-Q. Zhao and J. Yang, "Hyperspectral image denoising via sparse representation and low-rank constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 296–308, Jan. 2015.

[27] L. Zhuang, L. Gao, B. Zhang, X. Fu, and J. M. Bioucas-Dias, "Hyperspectral image denoising and anomaly detection based on low-rank and sparse representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[28] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.

[29] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[30] Q. Zhang, Q. Yuan, J. Li, X. Liu, H. Shen, and L. Zhang, "Hybrid noise removal in hyperspectral imagery with a spatial–spectral gradient network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7317–7329, Oct. 2019.

[31] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[32] M. Li, Y. Fu, and Y. Zhang, "Spatial–spectral transformer for hyperspectral image denoising," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, vol. 37, no. 1, pp. 1368–1376.

[33] M. Li, J. Liu, Y. Fu, Y. Zhang, and D. Dou, "Spectral enhanced rectangle transformer for hyperspectral image denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5805–5814.

[34] M. Li et al., "Latent diffusion enhanced rectangle transformer for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 1, pp. 549–564, Jan. 2025.

[35] X. Tan, M. Shao, Y. Qiao, T. Liu, and X. Cao, "Low-rank prompt-guided transformer for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520815.

[36] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[37] G. Fu, F. Xiong, J. Lu, and J. Zhou, "SSUMamba: Spatial–spectral selective state space model for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527714.

[38] F. Xiong, J. Zhou, J. Zhou, J. Lu, and Y. Qian, "Deep parameterized neural networks for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525015.

[39] Q. Qian and T. Mallick, "Wavelet-inspired multiscale graph convolutional recurrent network for traffic forecasting," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 5680–5684.

[40] F. Xiong, J. Zhou, Q. Zhao, J. Lu, and Y. Qian, "MAC-Net: Model-aided nonlocal neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519414.

[41] Q. Zhang, Y. Dong, Q. Yuan, M. Song, and H. Yu, "Combined deep priors with low-rank tensor factorization for hyperspectral image restoration," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[42] Y. Chen, H. Zhang, Y. Wang, Y. Yang, and J. Wu, "Flex-DLD: Deep low-rank decomposition model with flexible priors for hyperspectral image denoising and restoration," *IEEE Trans. Image Process.*, vol. 33, pp. 1211–1226, 2024.

[43] L. Zhuang, M. K. Ng, L. Gao, J. R. Michalski, and Z. Wang, "Eigenimage2Eigenimage (E2E): A self-supervised deep learning network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16262–16276, Nov. 2023.

[44] T. Bodrito, A. Zouaoui, J. Chanussot, and J. Mairal, "A trainable spectral–spatial sparse coding model for hyperspectral image restoration," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 5430–5442.

[45] F. Xiong, J. Zhou, S. Tao, J. Lu, J. Zhou, and Y. Qian, "SMDS-Net: Model guided spectral–spatial network for hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 5469–5483, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.

[46] M. Li, Y. Fu, J. Liu, and Y. Zhang, "Pixel adaptive deep unfolding transformer for hyperspectral image reconstruction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 12913–12922.
[47] F. Xiong, J. Zhou, J. Zhou, J. Lu, and Y. Qian, "Multitask sparse representation model-inspired network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518515.
[48] Y. Li, J. Li, J. He, X. Liu, and Q. Yuan, "An optimization-driven network with knowledge prior injection for HSI denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531717.
[49] S. Bai, J. Z. Kolter, and V. Koltun, "Trellis networks for sequence modeling," 2018, *arXiv:1810.06682*.
[50] M. Dehghani, S. Gouws, O. Vinyals, J. Uszkoreit, and L. Kaiser, "Universal transformers," 2018, *arXiv:1807.03819*.
[51] E. Winston and J. Z. Kolter, "Monotone operator equilibrium networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 10718–10728.
[52] S. Bai, V. Koltun, and J. Zico Kolter, "Stabilizing equilibrium models by Jacobian regularization," 2021, *arXiv:2106.14342*.
[53] Z. Geng, X. Zhang, S. Bai, Y. Wang, and Z. Lin, "On training implicit models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 24247–24260.
[54] A. Gkillas, D. Ampeliotis, and K. Berberidis, "Connections between deep equilibrium and sparse representation models with application to hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 32, pp. 1513–1528, 2023.
[55] S. Bai, V. Koltun, and J. Z. Kolter, "Multiscale deep equilibrium models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 5238–5250.
[56] Z. Yu et al., "Searching central difference convolutional networks for face anti-spoofing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5294–5304.
[57] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–133, Jan. 2013.
[58] M. Ye, Y. Qian, and J. Zhou, "Multitask sparse nonnegative matrix factorization for joint spectral–spatial hyperspectral imagery denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2621–2639, May 2015.
[59] Y. Chang, L. Yan, and S. Zhong, "Hyper-Laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5901–5909.
[60] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.
[61] L. Pang, W. Gu, and X. Cao, "TRQ3DNet: A 3D quasi-recurrent and transformer based network for hyperspectral image denoising," *Remote Sens.*, vol. 14, no. 18, p. 4598, Sep. 2022.
[62] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.
[63] P.-W. Tang, C.-H. Lin, J.-K. Huang, and A. R. Huete, "A quantum-empowered SPEI drought forecasting algorithm using spatially aware mamba network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4705118.
[64] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024, pp. 103031–103063.



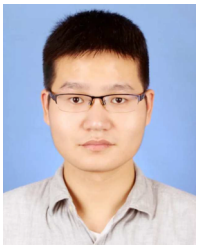
Jin Ye received the B.E. degree in software engineering from Nanjing University of Science and Technology, Nanjing, China, in 2023, where he is currently pursuing the M.S. degree.

His research interests include machine learning and hyperspectral low-level image processing.



Jingran Wang is currently pursuing the bachelor's degree in software engineering with Nanjing University of Science and Technology, Nanjing, China.

His research interests include machine learning and hyperspectral imaging.



Fengchao Xiong (Member, IEEE) received the B.E. degree in software engineering from Shandong University, Jinan, China, in 2014, and the Ph.D. degree from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2019.

He visited Wuhan University, Wuhan, China, from 2011 to 2012; Griffith University, Nathan, QLD, Australia, from 2017 to 2018; and the University of Macau, Macau, China, from 2021 to 2023. He is currently an Associate Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. His research interests include hyperspectral image processing, machine learning, and pattern recognition.

Dr. Xiong serves as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Jingzhou Chen (Member, IEEE) received the B.E. degree in computer science and technology from Sichuan University, Chengdu, Sichuan, China, in 2016, and the Ph.D. degree in computer science and technology from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2022.

From 2022 to 2023, he was a Senior Computer Vision Engineer with the Security and Risk Management Group, Ant Group, Hangzhou, where his work aimed to manage the risk from the image content. He is currently an Assistant Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China. His research interests include machine learning, computer vision, and remote sensing image analysis.



Yuntao Qian (Senior Member, IEEE) received the B.E. and M.E. degrees in automatic control from Xi'an Jiaotong University, Xi'an, China, in 1989 and 1992, respectively, and the Ph.D. degree in signal processing from Xidian University, Xi'an, in 1996.

From 1996 to 1998, he was a Post-Doctoral Fellow with Northwestern Polytechnical University, Xi'an. Since 1998, he has been with the College of Computer Science, Zhejiang University, Hangzhou, China, where he became a Professor in 2002. From 1999 to 2001, 2006, 2010, 2013, 2015 to 2016, and 2018, he was a Visiting Professor with Concordia University Montreal, QC, Canada; Hong Kong Baptist University, Hong Kong; Carnegie Mellon University, Pittsburgh, PA, USA, Canberra Research Laboratory, NICTA, Canberra, ACT, Australia; University of Macau, Macau, China; and Griffith University, Nathan, QLD, Australia, respectively. His research interests include machine learning, signal and image processing, pattern recognition, and hyperspectral imaging.

Dr. Qian is currently an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:37:31 UTC from IEEE Xplore. Restrictions apply.